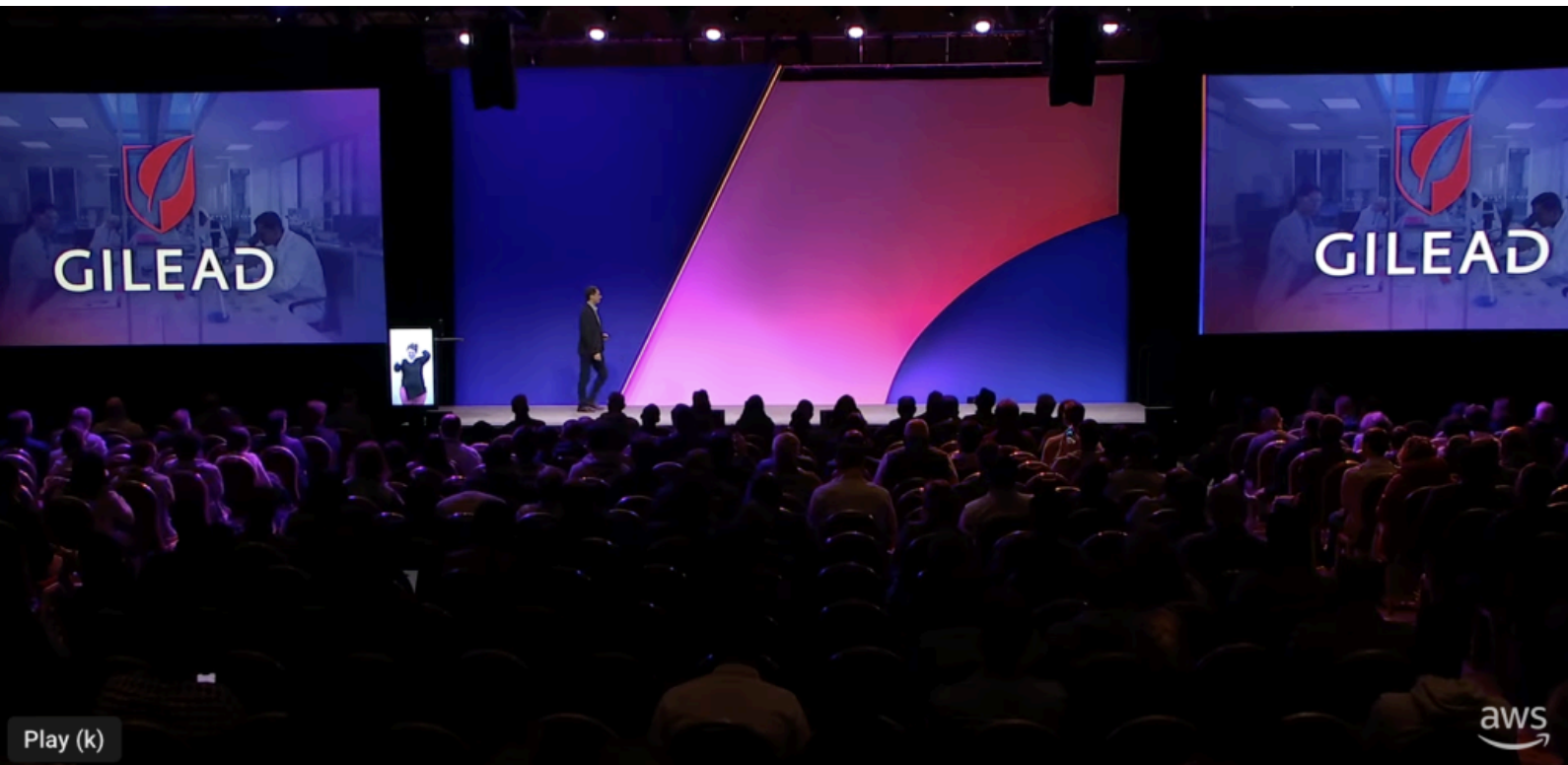


Key takeaways from Gilead's Innovation Talk at re:Invent 2023

by Dan Sheeran | on 06 MAR 2024 | in [Generative AI](#), [Industries](#), [Life Sciences](#) | [Permalink](#) | [Share](#)

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At [re:Invent 2023](#), Marc Berson, Chief Information Officer of [Gilead Sciences](#), joined an Innovation Talk on '[Generative AI for Industries](#)', along with Shaown Nandi, AWS Director of Technology for Industries. The talk highlighted Gilead's approach towards using generative AI for accelerating therapeutic breakthroughs, and emphasized their commitment to establishing a resilient cloud infrastructure and solid data foundations, to enable the technology at an enterprise level.

Here are the key takeaways.

Accelerating Therapeutic Advancements with Generative AI

Pharmaceutical companies face significant pressure to stay at the forefront of scientific innovation and expedite the delivery of groundbreaking therapeutics to patients grappling with life-threatening diseases. However, the process of developing a new drug typically spans over a decade, can involve billions of dollars, and faces a daunting 90% failure rate before it can reach the commercial market.

Generative AI can transform drug discovery by speeding up the identification and design of novel compounds and optimizing therapeutic development processes. The new technology can further shorten cycle times, by automating manual and redundant processes related to regulatory submissions, market access, and commercialization.

Gilead is at the forefront of harnessing the transformative potential of Generative AI to accelerate scientific advancements, and deliver innovations leveraging the confluence of cloud, data, and AI. The organization is exploring generative AI solutions for a range of high-ROI use cases – for augmenting decision making, increasing efficiencies, and accelerating insights from cloud-scale data.

"Navigating the industry's complex challenges requires companies to be agile, innovative, and adaptive to the evolving landscape—we need to seek every advantage we can get. We're exploring Generative AI with an enterprise mindset,

implementing an enterprise approach to identify high value gen AI use cases for investment.”

– Marc Berson, CIO, Gilead

Transforming Drug Discovery: Reimagining Target Assessment with Generative AI

Gilead is investing in a range of generative AI use cases to enhance its R&D capabilities, one use case of high interest being **target assessment**.

Target assessment is a crucial step in early stage drug discovery that involves evaluating and characterizing biological molecules (proteins or nucleic acids) for targeting by potential therapeutic agents. A thorough assessment helps researchers prioritize which targets are worth pursuing for potential new medications with a higher likelihood of success. However, the process involves mining petabytes of data, identifying relevant research, analyzing that research, and applying the research insights at speed.

Gilead is leveraging generative AI to bring step changes in the efficiency and effectiveness of target assessment process, using specialized LLMs (Large Language Models) for querying and analyzing vast amounts of scientific literature and data bases, at speed. This helps scientists perform natural language searches to summarize research literature, simulate molecular interactions with simple queries, and dynamically produce and update synthesized research reports, including references – ultimately accelerating the development of safer and more effective medicines.

Reimagining Target Assessment in drug discovery using generative AI



Use LLM tools to quickly analyze vast amounts of scientific literature and databases



Identify relevant research related to potential drug targets



Significantly speed up the initial stages of target identification and validation



“It’s not just about technology, it’s about transforming the way we work.”

Gilead envisions a substantial reduction in the time needed for target assessment – by several months – thanks to the efficient generation of high-quality target assessment reports facilitated by generative AI. The company is also exploring the reuse of these LLMs, architecture patterns, and LLMOps (Large language Model Operations) across other use cases across the company. *“The idea is to essentially do more with the same amount of resource,”* says Berson.

Generative AI can transform the way research scientists work

CAPABILITIES

- Interact using natural language
- Produces standardized research reports
- Generates quality results with speed

BENEFITS

- Accelerates delivery of summarized content
- Produces reusable architecture patterns and LLMops
- Potential for re-use across other areas



Establishing Robust Cloud & Data Foundations for Unlocking Generative AI

Despite the buzz surrounding models, Gilead recognized early on that the key to differentiated and potent generative AI applications lies in implementing a comprehensive end-to-end data strategy. A robust cloud infrastructure, built on AWS, enables Gilead to scale its operations efficiently, and to innovate with speed and agility. The organization has [moved 80% of its compute and storage workloads to AWS](#), and now runs its mission critical workloads on the cloud. *"We value the resilience, speed, and scalability that cloud provides," says Berson. "It is a strategic business priority for Gilead, and imperative to transforming our entire value chain, from drug discovery to commercialization"*

"At Gilead, we are building ML and generative AI on top of a solid cloud, data, and analytics foundation, built on AWS. Additionally, we have automated MLOps to realize efficiencies throughout the ML lifecycle."

[Gilead's enterprise data platform](#) enables the [harnessing of ML and generative AI capabilities](#), without the need to reinvent the wheel. Operating on a **data mesh architecture**, the platform centralizes the organization's data, scales its management, and fosters collaboration throughout the organization. And, a **decentralized data ownership** model gives control of the data to business teams that are the closest to it and possess contextual understanding.

The other distinctive feature of Gilead's data mesh is its '**data-as-a-product**' approach. Data producers across the organization have published over 300 data products on the enterprise data platform. These are readily available for use by data consumers, and utmost care is taken to ensure that the data is discoverable, addressable, understandable, trustworthy, useful, accessible, secure, and interoperable. Furthermore, a range of self-service data solutions to manage the full lifecycle of these data products support the strategy, by reducing the overall cost of decentralized data ownership.

"With AWS, we are implementing modern data organization and engineering practices, moving away from traditional, monolithic approaches, to respond more quickly to evolving business needs,"

As a global life sciences organization handling sensitive patient data, Gilead fortifies data security and compliance with a powerful data governance operating model, grounded in a federated decision-making accountability structure. "We have central standards and governance, but federated domain ownership," says Berson. "This allows us to respond to change with agility, and increase ROI and sustainability in the face of growth."

With this robust foundation in place, Gilead is expanding the use of AI, machine learning, deep learning, and analytics as an enterprise-wide capability – transforming complex data into valuable insights, and uncovering hidden relationships among molecules, targets, and diseases. And, it is collaborating with AWS for providing its data scientists with the latest advances in technology to deliver ground-breaking therapeutic innovations, towards its bold vision of creating a healthier world for all.

“We chose AWS—not just because of their technology, but for their passion for transforming the industry and inventing together. We are truly seeing the art of the possible.”

Future-proofing: Moving Generative AI Use Cases from Prototype to Production

The next 3 to 5 years will be pivotal in demonstrating the value of generative AI in drug discovery, as the industry transitions to more complex and targeted therapies. As more use cases progress from pilot to prototyping to production, Gilead is establishing guiding principles for the pragmatic use of generative AI – with a focus on ensuring responsible usage – to future-proof itself.

Berson closed the session with some thought-provoking advice for organizations jumping on the generative AI bandwagon – establish your own set of guiding principles, and get executive engagement and buy-in right at the outset. *“Make sure generative AI does not become an IT project, but something that is built in close collaboration with the business, measuring success with business outcomes,”* he cautioned.

[Watch a replay of this Innovation Talk on YouTube.](#)

To learn about other announcements and highlights for HCLS at re:Invent, [check out this blog](#).